

Doctoral Progress Review and Research Plan

Laser-Controlled Breakdown in Atmospheric-Pressure High-Voltage Gaps: Plasma-Channel Diagnostics, Stochastic Switching Dynamics, and Predictive Modelling

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Professional discussion after two years of doctoral study

Review draft - 16 September 2026

Application context: transition toward SF₆-free and, where feasible, F-gas-free high-voltage switching.

Tier 1 evidence: laser-triggered closure of an atmospheric-air gap, stochastic prediction and bounded TCO.

Claim boundary: no demonstration of complete switchgear replacement, current interruption or dielectric recovery.

Submission target: August 2028. Tier 2 and Tier 3 remain separate follow-on research.

Reading guide

The review distinguishes historical evidence, proposed doctoral contributions and subsequent research. It asks the opponent to assess scope and feasibility; it does not claim that planned results or formal approvals already exist.

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References

Evidence boundary. Seven voltage-reduction series originate in May 2024 Master's work. Three additional delay series originate in February 2025. The general irradiance-delay trend remains to be verified and is not assumed in the proposed analysis.

1. Purpose and proposed dissertation scope

Candidate: Ing. Michal Sakala
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Supervising department: Department of Economics, Management and Humanities (K13116), Faculty of Electrical Engineering, Czech Technical University in Prague
Experimental collaboration: HiLASE Centre, Institute of Physics of the Czech Academy of Sciences
Review date: 16 September 2026
Study start used for planning: 1 September 2024
Target: dissertation submission in August 2028; the defence date is a separate administrative milestone.
This study supports the professional discussion after the first two years of doctoral study. Its purpose is to assess the research completed so far and determine whether the proposed scope, methods and resources provide a credible route to a doctoral contribution. It is a research proposal for assessment by the supervisor and board, not a record of their approval. The document is self-contained: its scientific argument does not require access to a personal knowledge base or accompanying planning documents.

Long-term application context: transition away from SF₆. The power sector is replacing SF₆ because of its climate impact and new European restrictions, using architectures such as vacuum interruption with clean air or pressurised alternative gases. Tier 1 does not claim to replace complete SF₆ switchgear. It studies laser-triggered closure of a bounded atmospheric-air gap, independently validates probability and delay, and evaluates TCO for a defined laboratory switching service. It does not test insulation coordination, short-circuit current interruption, post-arc dielectric recovery or product lifetime. Any relevance to SF₆ substitution remains conditional on later validation for a defined switching function, medium, voltage/current rating and duty cycle. Tier 2 and Tier 3 are separately resourced follow-on research outside mandatory completion.

Context sources: Regulation (EU) 2024/573, Article 13; CIGRE WG A3.41, Technical Brochure 871 (2022).
Proposed English title: Laser-Controlled Breakdown in Atmospheric-Pressure High-Voltage Gaps: Plasma-Channel Diagnostics, Stochastic Switching Dynamics, and Predictive Modelling.

Proposed Czech title: Laserem řízený průraz atmosférických vysokonapěťových mezer: diagnostika stavu plazmového kanálu, statistická dynamika spínání a prediktivní modelování.

The selected research programme is a three-tier structure. Only Tier 1 is mandatory. Tier 1 connects calibrated experiments in atmospheric air, stochastic prediction and an economic decision model for a defined laboratory switching service. Tier 2 concerns subsequent transfer to CO₂ and controlled pressure. Tier 3 concerns a later diagnostic, control or application demonstrator. Neither Tier 2 nor Tier 3 is required for the dissertation, its mandatory publications or submission in August 2028.

Tier	Scientific scope	Relation to dissertation completion
1: mandatory	Atmospheric air; timing metrology; channel descriptors; probability and delay; independent validation; bounded TCO	Complete for submission in August 2028
2: follow-on	Transfer to CO ₂ and a bounded pressure domain; medium-specific calibration	Separate project; no mandatory thesis claim
3: follow-on application	One later diagnostic, recovery or adaptive operating study	Separate resources and decision; no mandatory thesis claim

The scope is bounded in experimental dimensions but deep in evidence. The primary condition is one confirmed atmospheric-air geometry and polarity; a second geometry or polarity may be added as a controlled robustness check if repeatability and resources permit. The core question is whether a measured channel descriptor improves prediction on new experimental sessions and whether those predictions support a less costly choice of operating conditions at a fixed technical requirement. A null predictive or economic result remains scientifically informative when the experiment has sufficient sensitivity and reports quantitative limits. Sufficiency for the degree remains a decision of the relevant academic bodies.

less costly choice of operating conditions at a fixed technical requirement. An outcome showing no useful predictive or economic improvement remains scientifically informative if the experiment has sufficient sensitivity and reports quantitative limits. Sufficiency for the degree remains a decision of the relevant academic bodies.

2. Research completed and provenance of the results

2.1 Experimental work before doctoral enrolment: May 2024

The initial seven measurement series were acquired in May 2024 as part of the candidate's Master's thesis work. They investigated the change in breakdown voltage with laser wavelength, power, focusing and pulse regime. These measurements form the experimental basis of the ICOLSE 2024 contribution, Analysis of Laser Parameters for Efficient Quantum Ionization in Air for HV Circuit Triggering [1]. They also appear as series 1-7 in the APL 2025 paper [2]. They must therefore be counted once as an experimental dataset, even though two publications discuss them.

These results are the starting competence and evidence inherited by the doctorate. They demonstrate experience in aligning an ultrafast laser with a high-voltage spherical gap, measuring breakdown and processing the dependence on optical settings. They are not presented as seven new experiments performed after doctoral enrolment. The acquisition chronology and reuse of the seven series were confirmed by the candidate for this review; the retained laboratory archive contains May 2024 and February 2025 acquisition folders.

2.2 New measurements during the doctorate: February 2025

Three additional series were acquired in February 2025 and incorporated as series 8-10 of the APL paper. These extend the investigation from a voltage-reduction study to the electrical delay between the laser-trigger marker and breakdown. They use three laser settings and different applied voltages. The retained archive includes sessions dated 21 and 24 February 2025, waveform records and analysis workbooks. This review does not claim to have reconstructed every published value from the raw waveforms.

The APL article combines the seven inherited series and these three new series. It is a published dissemination result with the candidate as first author [2]. The conference took place in June 2025; the new measurements therefore belong to the first doctoral study year, rather than to 2025/2026 merely because the article is discussed in a later progress report.

2.3 Consolidation and planning in 2025/2026

The materials prepared for the current discussion develop the next research phase: calibrated optical time zero, explicit definitions of failed shots, reproducible data records, channel diagnostics, reduced modelling and validation on unseen sessions. These are methodological proposals and preparation activities. They do not establish that the new confirmatory dataset, validated predictive model or TCO results already exist.

COMSOL modelling and a parametric study are mentioned in working records. This dossier treats the electrostatic and reduced-model tasks as work requiring documented configuration, outputs and validation; it does not assert a completed or experimentally validated simulation result without that evidence. The review of course completion, internationalisation and publication recognition is kept separate from this scientific progress account.

Period	Activity	Evidence status and interpretation
May 2024	Seven voltage-reduction series for Master's work	Prior experimental foundation, reused in ICOLSE and APL
2024	ICOLSE contribution	Conference dissemination of the prior dataset [1]
February 2025	Three additional delay series	New doctoral experimental work, reported in APL [2]
2025	APL publication	Combined account of inherited and new measurements; not ten newly acquired doctoral series
2025/2026	Research synthesis and preparation of W	Proposed methods and scope; no new high-statistics result claimed here
September 2026	Professional-discussion preparation	Decision on scope, evidence standard and route to August 2028

3. What the existing experiments establish

3.1 Apparatus and tested domain

The APL paper describes stainless-steel spherical electrodes of 20 mm diameter, gaps of 5 and 10 mm in atmospheric air, a Technix SR 40 kV-5 kW DC source and a picosecond PERLA laser. The reported nominal pulse duration is 1.2 ps and the repetition rate is 1 kHz. The tested wavelengths are 1030, 515 and 257.5 nm. Voltage and current were recorded using a Tektronix MDO4054B oscilloscope, a Tektronix P6015A probe and a Pearson 1423 current monitor. The paper reports at least five repetitions per configuration [2]. These are descriptions of the historical experiment, not a certification of the equipment currently available for the next campaign.

Series	Acquisition	Wavelength	Gap	Principal quantity
1-2	May 2024	1030 nm	5 and 10 mm	Breakdown voltage
3-4	May 2024	515 nm	5 and 10 mm	Breakdown voltage
5-6	May 2024	257.5 nm	5 and 10 mm	Breakdown voltage
7	May 2024	1030 nm	5 mm	Breakdown voltage, single-pulse regime
8-10	February 2025	1030 nm	5 mm	Electrical delay at three laser settings

3.2 Reported voltage and delay results

The maximum relative voltage reductions tabulated in APL range from approximately 18% to 31%. For the 515 nm, 10 mm configuration, the tabulated reference and laser-assisted voltages are 29.5 and 20.3 kV, respectively, corresponding to approximately 31%. These values are campaign observations; the minima must not be relabelled as a calibrated 50% breakdown-voltage curve or as an operating reliability guarantee [2].

The delay table contains values from 0.164 to 8.124 μs . Within the reported series, decreasing applied voltage is associated with longer delay. However, the delay is referenced to an electrical laser-trigger signal. It is not yet the calibrated interval between optical pulse arrival in the gap and electrical breakdown [2].

The general claim that increasing laser irradiance always shortens delay is not adopted. The table and narrative in the local APL material require reconciliation on this point. No irradiance ordering is used to infer an optimum, interpolate a monotonic response or calculate economic benefit. The three settings are retained as reported operating conditions pending verification of their calibration and assignment. This open issue does not invalidate the proposal to measure the response prospectively using a calibrated protocol.

3.3 Limits relevant to the next phase

The historical campaigns demonstrate laser-assisted breakdown and provide a useful voltage and timing domain for planning. They do not yet demonstrate a shot-resolved channel-state predictor, transfer across independently realigned sessions, rare-event reliability, de-embedded physical jitter or an economic optimum. The small repetition count is also inadequate for strong statements about the tails of the delay distribution.

There is no measured product-level evidence here for longer electrode life, avoided network outages, short-circuit interruption, reduced maintenance of industrial switchgear or replacement of SF_6 . Such benefits will not enter the base-case TCO as achieved savings. The future economic contribution evaluates a bounded laboratory service using measured performance and explicit cost scenarios.

4. Physical foundations, literature and research gap

4.1 Breakdown and the laser-created channel

Streamer inception and propagation involve local electric-field enhancement, charge transport and ionisation. The reduced electric field E/N connects electrical forcing to gas number density, but a single reduced-field value does not describe the full geometry or discharge history. Streamer physics, diagnostics and simulation methods provide the physical basis for selecting measurements and interpreting model limits [3].

A laser-created channel may provide initial ionisation and may subsequently alter the gas state through deposited energy. The relative contributions of free-charge dynamics and a longer-lived density perturbation must be treated as hypotheses for the studied conditions. Bright emission is not itself an electron-density, gas-density or temperature measurement. A channel descriptor can be useful empirically without being a direct measurement of one of those physical quantities; the dissertation must state that distinction.

The proposed work therefore begins with one demonstrably repeatable descriptor of geometry or temporal optical response. It adds a calibrated density-sensitive diagnostic only if available apparatus and an identifiable forward model justify that claim. Full plasma chemistry is not a mandatory deliverable. Electrostatic calculation for the actual electrode geometry and a reduced physical interpretation are sufficient modelling starting points.

4.2 Stochastic response and measurement uncertainty

At a fixed nominal setting, breakdown may occur at different delays or not occur within the observation window. Analysis must retain attempted shots, successful events, delayed events and acquisition failures as distinct categories. A shot without breakdown by the end of the observation window is a right-censored delay observation. Survival methods provide a basis for representing incomplete event times [4]. They do not justify treating an acquisition failure as a physically observed non-event.

Timing offsets, bandwidth, threshold detection and between-session variation enter the inference. Measurement models and propagation of uncertainty follow the principles of the GUM and its Monte Carlo supplement [5,6]. The uncertainty budget must include shared terms and correlations. Subtracting an assumed instrument variance from an observed jitter variance is justified only under the relevant independence and calibration assumptions.

4.3 From prediction to a technical and economic decision

Predictive accuracy and economic usefulness are different outcomes. A descriptor that improves a statistical score may have little value if it does not change the admissible operating choice, or if its calibration and acquisition costs exceed the resulting saving. Conversely, a lower-energy setting may be more expensive per compliant operation if it causes more unsuccessful attempts or longer resetting time.

Life-cycle costing provides the basis for comparing acquisition, operation, replacement and residual value over a common period. Sensitivity analysis is necessary when utilisation and service life are uncertain [7]. In this dissertation, that method is adapted to a laboratory switching service; building-specific assumptions or foreign discount rates are not transferred automatically.

4.4 Bounded originality claim

The proposed contribution is the experimentally traceable connection between channel observables, stochastic breakdown outcomes and cost-constrained operating decisions in a bounded family of atmospheric-air configurations. One reference geometry and polarity form the primary condition; a second geometry or polarity may test robustness when the core evidence is secure. The project does not claim the first laser-triggered gap, first survival model or first TCO analysis. Its originality must be established against relevant combinations of these methods, not by the presence of any method alone.

Before the confirmatory campaign, the literature comparison will explicitly record medium, pressure, pulse regime, timing reference, channel observable, treatment of failed shots, independent validation and cost boundary. This targeted comparison will refine the novelty statement. The present review provides a justified research direction, not a completed proof of worldwide novelty.

5. Research questions and intended contributions

Central question: Can measurements of a laser-created channel improve prediction of breakdown probability and timing in atmospheric air, and support selection of the least-cost operating conditions that satisfy a predeclared technical requirement?

Question	Test	Interpretation of a negative result
RQ1: Is physical delay resolved by the measurement chain?	Optical time-zero calibration and uncertainty budget	Limit the timing claim to the available resolution
RQ2: Does channel state add predictive information?	Compare baseline M0 and channel-informed M1 on untouched sessions	Bound the gain and retain the simpler model

Question	Test	Interpretation of a negative result
RQ3: Does the prediction support admissible operating choices?	Evaluate a frozen choice rule on independent validation data	Identify conditions where the rule fails or cannot certify performance
RQ4: Which admissible condition has the lowest cost?	TCO and cost per compliant operation with sensitivity analysis	Report no stable optimum or no cost advantage within uncertainty

Here RQ denotes a research question, H a predeclared testable hypothesis, and P a research proposition whose scope and limits are evaluated. The identifiers are independent: one question may motivate several statements. The current evidence register contains H1-H4 and P1-P3.

Statement	Planned test and interpretation
H1: laser excitation changes gated breakdown probability	Matched no-laser comparison at fixed normalised voltage; report the effect or a quantitative bound
P1: delay distributions can be characterised within the measured domain	Calibrated timing and a two-sided analysis accounting for voltage, session and electrode state; report unresolved trends as limits
H2: channel descriptors improve held-out prediction	Frozen M0/M1 comparison on independent sessions; report gain or an upper bound
H3: polarity or geometry modifies a channel-breakdown relationship	Conditional robustness test only if the core evidence is repeatable; a null result bounds the interaction
H4: a reduced model predicts a core observable on independent sessions	Test against a declared tolerance without adding unconstrained states after validation
P2: optical and electrical event ordering is identifiable within diagnostic resolution	Synchronized measurements and independent repeat; restrict physical stage labels to supported observations
P3: measured performance and costs delimit technically admissible operating choices	Independent technical validation and TCO sensitivity; report an empty or unstable feasible set when indicated

H3 is supporting rather than a completion condition. P1-P3 make no assumption that effects are monotonic, that physical stages are always identifiable, or that an economically favourable operating point exists.

Contribution	Mandatory evidence	Completion meaning
C1: atmospheric operating domain	Calibrated conditions, attempted-shot records, uncertainty and independent repetition	A bounded, reproducible probability and timing map
C2: predictive assessment	Frozen M0/M1 comparison and held-out sessions	Quantified benefit or defensible upper bound on benefit
C3: techno-economic decision method	Defined service, traceable costs, feasibility constraints and uncertainty	Reproducible cost comparison and break-even conditions
C4: reproducible evidence	Versioned data, calibration, analysis and provenance	Every reported result traceable to its supporting records

The contribution sequence follows expected completion: C1 and C2 precede C3; C4 closes the traceable Tier 1 evidence package after TCO synthesis. C5 and C6 are subsequent CO₂/pressure-transfer contributions in Tier 2, subject to a separate protocol and resources. Their internal order will follow the approved Tier 2 schedule. They are not conditions for Tier 1 completion. The selected scope and the adequacy of C3 for the candidate's programme require academic assessment at the discussion. The departmental context motivates this integration, but does not by itself certify programme compliance [10].

6. Experimental design and metrology for Tier 1

6.1 Controlled scope

Use atmospheric air only. Record ambient pressure, temperature and humidity rather than assuming constant density. Begin with one reference electrode geometry, polarity and confirmed laser branch. Add one second geometry or polarity as a controlled robustness check only when the primary result is repeatable, the comparison is powered and it fits the frozen workload. An optical condition may be added only when it tests transfer rather than opening a new parameter sweep. Do not perform a full factorial sweep over all wavelengths, focus positions, voltages, polarities and pulse regimes.

The initial sequence comprises no-laser reference measurements, a small optical/voltage screening design and a reduced confirmatory matrix. A near-threshold region is informative for probability estimation; highly successful operating points may be added for the defined application requirement. The robustness check is subordinate to the primary atmospheric evidence and may be dropped at the June stop/go gate. Controls and safe working limits are set by the actual apparatus and laboratory owner.

6.2 Operational quantities

An attempted shot is a trigger request made in the declared ready state. Breakdown is identified by a predeclared electrical criterion based on voltage and/or current. Delay is the interval from optical arrival at the interaction region to that criterion. The observation window and any early-event exclusion are fixed before confirmation. A compliant operation must meet the prescribed timing and electrical conditions, not merely produce visible light.

Normalized voltage $k = U/U_{50}$ uses a contemporaneous no-laser reference obtained by a defined procedure and observation time. U_{50} is an estimated quantity with uncertainty. Electrode conditioning, ramp or dwell protocol and environment must be reported. Historical minima are not substitutes for this reference.

Channel features must be extracted from a defined time interval before breakdown. A feature containing the breakdown flash would leak the outcome into the predictor. If a feature is available only after a fixed landmark time, the model must condition on survival to that landmark and report coverage of early events. Failed shots must have the same planned acquisition opportunity. Non-shot-resolved diagnostics may support an ensemble interpretation, but not a claim of individual-shot prediction.

6.3 Measurement chain

Quantity	Method	Required check
Applied voltage/current	Calibrated probes and common acquisition clock	Bandwidth, offset, saturation and marker sensitivity
Optical time zero	Fast optical pick-off with known path relationship to the gap	Relative channel delays and pulse-arrival uncertainty
Laser input	Energy and beam measurement in the active configuration	Distinguish setpoint, delivered pulse energy and electrical consumption
Channel descriptor	Synchronized optical signal or calibrated image	Spatial/time calibration, repeatability and absence of outcome leakage
Environment/electrodes	Ambient sensors and configuration/history log	Link records to session, block and attempted shot
Economic inputs	Equipment records, power/time measurements and service information	Cost date, allocation rule, scope and uncertainty

The measured delay is corrected by a calibrated net offset, with uncertainty propagated from optical path, acquisition channels, detector response and the event-marker algorithm. Pulse-to-pulse fluctuation and uncertain fixed offsets are distinguished. Where the physical variation cannot be resolved, the report gives a bound rather than claiming a de-embedded jitter value.

6.4 Sample size and independent repetition

The pilot determines event rates, between-session variability and the feasible number of conditions. Confirmatory sample size follows a declared interval-width or power target, with session clustering included. There is no universal five-shot or hundred-shot requirement. For perspective, even 100 independent successes without a failure imply only about a 0.9705 one-sided 95% lower confidence bound on success probability; they do not establish 99.9% reliability. This follows from $p_{\text{lower}} = 0.05^{1/n}$ under the independent binomial assumption.

Block by day and electrode history, randomize settings within safe blocks and repeat the principal comparisons in independently realigned sessions. Hold out complete sessions rather than randomly mixing nearby shots into training and validation. The final sample count, stopping rule and effect-size threshold are frozen after the pilot. Repeated inspection must not become an undeclared stopping rule.

7. Prediction, validation and decision rules

M0 uses normalized voltage and measured laser input, with declared environment and session terms where appropriate. M1 adds the predeclared channel descriptor using the same training and validation partitions. Compare models of proportionate complexity so that an apparent gain cannot be attributed simply to an unrestricted increase in flexibility.

For probability, use a binomial model and report calibration as well as a proper scoring rule such as the Brier score. For delay, use a survival or accelerated-failure-time model that retains right-censored records; assess prediction of a predeclared time quantile or survival probability. Report timing distributions conditional on the stated population. Jitter among successful events is not an unconditional description of all attempted shots.

Model selection and preprocessing use training data only. Independent validation data remain untouched until the analysis and choice rule are frozen. Uncertainty is assessed at the session level where the sampling structure requires it. Report performance by condition as well as a pooled score, with the operating domain and failed predictions visible.

There are two prediction times. A pre-trigger operating choice can use settings and information from previous observations. A channel descriptor measured after the current laser pulse may improve prediction of that shot's subsequent breakdown, but cannot retrospectively choose the energy of that pulse. Tier 1 therefore evaluates an offline operating rule using repeated data and calibrated configuration-level response. It does not claim real-time control of the same shot. Repeated online adaptation or a controller belongs to Tier 3.

Freeze the recommended setting using training/calibration data and the declared cost scenario. Test its technical feasibility on a separate session. Selecting the cheapest condition on the final validation results and reporting those same results as independent confirmation is prohibited. If the best choice changes materially across plausible cost scenarios, report the alternatives and the information needed to distinguish them.

7.1 Role and execution of the theoretical model

The theoretical work starts with an electrostatic field calculation for the measured electrode geometry and applied voltage. Record geometry, boundary conditions and numerical convergence, then use the resulting field distribution to interpret the tested operating domain. This is an input calculation, not a simulation proving that breakdown will occur.

Next, construct a reduced description of channel evolution using only states that the diagnostics can constrain. Compare plausible charge-loss and gas-density timescales with the observed timing; distinguish measured inputs, literature parameters and fitted parameters. An optical descriptor may remain an empirical proxy if conversion to a physical state is not identifiable. A full three-dimensional plasma-chemistry model is not required.

Connect the reduced description to the probability or survival model and compare it with M0 on the same independent sessions. A proposed mechanism must earn its place through identifiable parameters, sensitivity and predictive performance. Report what the data cannot distinguish. The modelling deliverables are a field map, an explicit reduced-model specification, a parameter/uncertainty table and an independent predictive comparison, rather than a promise to reconstruct every discharge mechanism.

For the economic operating choice, use the response available before selecting a setting. If M1 uses a channel feature measured after the laser pulse, integrate over that feature's configuration-dependent distribution before predicting the cost or compliant-operation probability of a future attempt. Charge recurring diagnostics only when the evaluated operating service uses them; document research and commissioning costs separately. This keeps shot-level prediction and configuration-level TCO consistent.

8. Mandatory techno-economic assessment

8.1 Decision, service and comparison boundary

The reference service is a scheduled, laser-assisted switching operation in the defined atmospheric-air laboratory gap, with a specified electrical test condition and allowed timing window. The decision is which validated laser/voltage configuration provides that service at the lowest cost. The initial comparison is among feasible configurations of the same architecture, not between unrelated laboratory gaps and industrial circuit breakers.

Define demand, timing interval, acceptable success probability, electrical duty and reset protocol with the supervisor and laboratory partner before cost ranking. Those numerical requirements are inputs to the

study, not values inferred from whichever configuration performs best. Report both total cost for the required demand and cost per compliant operation. A setting that cannot satisfy demand or performance is infeasible even if its nominal cost per attempt is low.

8.2 Two ownership scenarios

Shared facility: allocate use of the existing laser and diagnostics by a documented access-price or resource-consumption rule. Do not charge the full acquisition cost and the facility access fee for the same service. Costs already incurred are distinguished from incremental decision costs.

Dedicated installation: estimate the investment and operation of a functionally specified standalone installation, using dated quotations or documented ranges where possible. This is a scenario until such an installation and its performance are established. Do not assume an inexpensive replacement laser will reproduce PERLA performance without evidence.

The two perspectives are reported separately. A low marginal cost at HiLASE is not proof of a commercially viable standalone device. Conversely, allocating the entire research laser to one experiment may overstate the cost of a shared service.

8.3 Cost model and data

For configuration x , horizon H and discount rate r , define discounted total cost as:

$$TCO_x = C_{0,x} + \sum_{t=1,\dots,H} (C_{op,x,t} + C_{rep,x,t} + C_{fail,x,t}) / (1+r)^t - R_{x,H} / (1+r)^H$$

C_0 is initial investment within the selected boundary; C_{op} covers operating resources and scheduled maintenance; C_{rep} covers replacements; C_{fail} covers incremental failure consequences not already included elsewhere; R is residual value. Use a consistent price year, currency and real or nominal discounting convention. Record VAT treatment according to the selected decision-maker. Depreciation is not added as a second cash expenditure to an already included acquisition cost.

Input	Preferred evidence	Treatment when unavailable
Laser, optics, diagnostics and installation	Dated records or quotations	Explicit investment range
Facility allocation and utilisation	Access terms, booked and productive time	Separate utilisation scenarios
Electricity and cooling	Measured system input in active/idle states	Documented equipment bounds; never optical energy alone
Operator and alignment time	Session time logs and labour-cost convention	Measured pilot distribution or labelled range
Maintenance and replacement	Service records and component history	Scenario range; no life-extension claim
Successful/timely operations	Independent experimental validation	Confidence intervals and unmet-constraint reporting
Reset after failure	Observed sequence, time and consumables	Explicit reset scenario without assumed automatic retries

The optical pulse energy measures experimental input to the gap. Wall-plug electricity, standby consumption, cooling and setup time determine the corresponding operating cost. Their relationship is measured or bounded separately.

8.4 Performance-normalized cost and constraints

Let $q(x)$ denote the probability of a compliant operation, including the specified timing requirement. With $A(x)$ attempted operations over the horizon, expected compliant output is $A(x)q(x)$ for a stationary scenario. A transparent diagnostic metric is:

$$c_{ok}(x) = TCO_x / [A(x) q(x)]$$

This ratio uses discounted costs and an undiscounted physical output count; that convention is stated explicitly. For fixed-demand comparison, $A(x)$ and the associated time and operating cost must be adjusted consistently to the demand and capacity model. The ratio alone is not a substitute for satisfying demand. If repeated attempts are allowed, success, reset time, capacity and dependence between attempts are modelled explicitly. Failure cost includes only additional consequences, avoiding double counting energy, operator time or lost production.

The feasible domain requires the chosen lower confidence bound on $q(x)$ to meet q_{req} , the timing requirement to hold, and the operating duty to remain within the validated apparatus limits. If no point qualifies, report no demonstrated feasible solution. Do not relax constraints after seeing the results to create a preferred operating point.

8.5 Uncertainty and economic contribution

Evaluate sensitivity to utilisation, investment allocation, diagnostic cost, electricity, labour, failure/reset time and replacement interval. Propagate measured performance uncertainty jointly with cost uncertainty while preserving known correlations. Use scenario ranges when a probability distribution lacks evidence. A Monte Carlo output does not make unsupported inputs empirical [6,7].

The required outputs are the feasible operating domain, cost ranking with uncertainty, break-even conditions and the inputs that most affect the decision. Add the incremental cost of diagnostics before claiming that M1 creates economic value. If the cost ranking is unstable, the result is the boundary between preferred choices and the value of better information, not a single precise payback period.

This contribution links physical performance to resource allocation and investment/operation choices relevant to K13116. It remains bounded to the service studied. Avoided grid outages, industrial asset-life extension, avoided SF_6 costs and field reliability are excluded from the base case because the required system evidence is outside Tier 1.

9. Work packages and completion gates

WP0-WP5 cover preparation, acquisition and prediction in the atmospheric-air study. WP6 integrates the TCO analysis, and WP7 covers writing and submission. "Experimental campaign" refers generally to the laboratory acquisition; it has no separate numbered series. Dates are planning windows, subject to laboratory access and academic approval.

Work package	Period	Required result and gate
WP0: scope and metrology	Sep-Dec 2026	Confirm configuration, safety, optical reference, data protocol, service definition and cost boundary

Work package	Period	Required result and gate
WP1: baseline	Jan-Feb 2027	Repeatable no-laser reference and environmental/electrode blocks
WP2: channel pilot	Mar-Apr 2027	At least one interpretable, repeatable descriptor or documented diagnostic limit
WP3: screening	May-Jun 2027	Compact confirmatory matrix, sample size, performance requirements and frozen analysis; June stop/go
WP4: confirmation	Jul-Oct 2027	Locked atmospheric dataset and independent session repeat; Paper 1 preparation
WP5: prediction	Nov 2027-Feb 2028	Held-out M0/M1 assessment and independently tested choice rule
WP6: TCO	Cost collection from WP0; synthesis Mar-Apr 2028	Traceable cost model, feasibility, sensitivity and break-even results; Paper 2
WP7: thesis	Draft continuously; full draft Jun 2028	Review, publication/administrative eligibility check and August 2028 submission

The June 2027 gate tests whether the apparatus resolves the target effect, the operating domain is repeatable and the descriptor can be linked to shots without leakage. If diagnostics fail, narrow the predictive question and quantify the diagnostic limit while retaining the calibrated operating-domain and economic comparison. Any change in the promised scientific contribution must be brought to the supervisor; a fallback is not an automatic guarantee of degree sufficiency.

The atmospheric evidence and core model are closed by February 2028, with TCO synthesis by April. No new core campaign is planned for July-August 2028. Writing proceeds during experimentation. Cost collection begins with the first campaign rather than becoming an unsupported final chapter.

9.1 WP0 - Make the experiment measurable (Sep-Dec 2026)

What to establish: the available apparatus can produce traceable, shot-linked optical and electrical records.

1. Confirm the actual laser branch, gap/circuit, sensor inventory, laboratory permissions and booking.
2. Verify electrical-trigger timing against optical pulse arrival at the gap; define the breakdown marker, failed-shot codes and uncertainty budget.
3. Freeze the shot record and begin logging equipment use, electrical consumption and operator/reset time for TCO.

Output: approved configuration and measurement protocol.

9.2 WP1 - Establish the no-laser reference (Jan-Feb 2027)

What to establish: how the selected atmospheric gap behaves without laser assistance.

1. Fix one reference geometry and polarity; document electrode history and ambient conditions.
2. Record both breakdown and no-breakdown attempts under a declared voltage/observation procedure, repeated across independent sessions.

3. Estimate the contemporary U_{50} curve and uncertainty; define $k=U/U_{50}$ for subsequent comparisons.

Output: reproducible baseline and its limits. **Theory in parallel:** calculate the static field in the actual geometry and test sensitivity to geometry, voltage and numerical mesh.

9.3 WP2 - Test whether the laser channel is observable (Mar-Apr 2027)

What to establish: whether a pre-breakdown channel feature can be measured reliably for each relevant shot.

1. Measure actual pulse energy, beam/focus properties and optical arrival time in the confirmed laser configuration.
2. Run a small laser-only and combined laser/HV pilot with synchronised waveforms and one candidate optical descriptor.
3. Repeat after realignment; determine whether the descriptor is available before breakdown and is more than detector noise or the breakdown flash.

Output: an interpretable descriptor **or** a quantified diagnostic limit. **Theory in parallel:** specify a minimal channel-evolution model using only identifiable states; keep an optical proxy distinct from direct electron density.

9.4 WP3 - Choose the decisive test (May-Jun 2027)

What to establish: a compact confirmatory design that can answer the dissertation question with available time.

1. Screen a small set of measured voltage/laser conditions; block by session and electrode history.
2. Use pilot event rates and session variation to determine attempt counts, primary contrasts and a stopping rule.
3. Freeze outcome definitions, M0/M1 inputs, channel-feature window and whole-session validation split. Hold a June supervisor stop/go review.

Output: dated decision and frozen confirmation protocol.

9.5 WP4 - Acquire new confirmatory evidence (Jul-Oct 2027)

What to establish: probability and delay distributions, with uncertainty, for the selected atmospheric conditions.

1. Execute the frozen matrix; retain every ready-state attempt, including no-breakdown and premature events.
2. Link optical timing, voltage/current, laser input, channel descriptor, environment and electrode history by shot ID.
3. Repeat principal contrasts in a separately initiated/realigned session and reserve complete sessions for later validation.

Output: QC-passed new dataset and Paper 1 evidence, clearly separated from the seven May 2024 and three February 2025 historical series. **Theory in parallel:** fit a baseline probability/delay model on training data and compare measured channel timescales with the reduced physical description.

9.6 WP5 - Test prediction on unseen sessions (Nov 2027-Feb 2028)

What to establish: whether channel information improves prediction beyond voltage and measured laser input, and whether a chosen setting remains technically feasible.

1. Freeze M0 (settings/context), M1 (M0 plus a predeclared pre-breakdown channel feature), preprocessing and scoring before opening reserved sessions.
2. Compare predicted breakdown probability and time-to-event distributions with complete held-out sessions; retain censored attempts and show failure regions.
3. Select an operating point from training/calibration evidence, then test that **preselected** point independently. A feature measured after a pulse cannot retrospectively select that pulse's energy.

Output: validated prediction or a documented lack of improvement, plus an independent technical check.

9.7 WP6 - TCO synthesis (Mar-Apr 2028)

What to establish: whether any validated configuration meets a predefined laboratory switching requirement at a defensible cost.

1. Consolidate costs logged since WP0: equipment use, electricity, operator/reset time, servicing and replacement. Distinguish research diagnostics from diagnostics required for routine operation.
2. Combine independently validated performance from WP5 with these costs; compare shared-facility and dedicated-installation scenarios separately.
3. Report feasible choices, uncertainty, sensitivity and break-even boundaries, or no demonstrated feasible/stable choice. Prepare Paper 2.

Output: bounded TCO and a defensible operating-choice result.

9.8 WP7 - Dissertation and submission (continuous writing; Jun-Aug 2028)

What to establish: a coherent account of new contributions, historical evidence and limitations, suitable for submission.

1. Write methods, results and limitations throughout WP0-WP5 and WP6.
2. Complete the full dissertation draft by June 2028; obtain review and check publication/study eligibility.
3. Revise and target submission in August 2028. The defence date is a separate administrative milestone.

Output: reviewed dissertation and submission package. No new core campaign is planned for July-August 2028; Tier 2 and Tier 3 remain outside mandatory completion.

10. Publications, study obligations and resources

10.1 Publication strategy independent of Tier 2

The existing dissemination record comprises ICOLSE 2024 [1] and APL 2025 [2], with the dataset overlap disclosed. Formal recognition is not assumed from a conference label or first-author position alone. The planned new papers use atmospheric-air evidence:

Paper 1, target submission October-November 2027: calibrated stochastic operation of the atmospheric gap, optical timing reference, uncertainty and independent repetition. Its contribution is the new confirmatory evidence, with explicit distinction from the 2024/2025 data.

Paper 2, target submission March-April 2028: predictive validation and techno-economic selection of operating conditions, including the limits of channel information and scenario sensitivity. If both papers use part of the same new dataset, that reuse is disclosed and their scientific questions are distinct.

At least the qualifying journal route must be selected early, with a fallback venue and allowance for review and revision. Submission is not acceptance. The older working target of December 2026 for an IEEE manuscript is not treated here as a new confirmatory-results paper; its identity and readiness remain to be reconciled with the supervisor. It is not counted as accepted or necessary evidence in this schedule.

10.2 Administrative planning boundary

The 2023 FEE Doctoral Study Code is the planning baseline. It requires completion of the study unit, including courses, a successful debate and a related accepted lead-author article; full-time completion is normally by the end of month 25. The State Doctoral Examination is required by the end of the third full-time study year. Before submission, the Code requires three related accepted/published articles, lead authorship of two and a qualifying venue for at least one lead-author article. Internationalisation and contribution statements also require attention [8].

For continuous full-time study starting in September 2024, the working schedule therefore targets study-unit closure in September 2026 and the examination in June 2027, ahead of the nominal August 2027 boundary. Individual dates, form of study and any exceptions must follow the official record. The earlier directive defines the role of the written study and discussion in determining the dissertation title and content [9]; its older credit-based wording is not substituted for the 2023 baseline.

The current dossier does not certify completed courses, publication recognition, the exact registered programme, an approved stay or a defence date. These must be reconciled with the ISP/KOS and the competent bodies. The submission target is August 2028, subject to meeting the requirements; it is not an assertion that a defence is already booked. Planning assumes a monograph; a dissertation assembled as a collection of publications requires a separate check of its additional conditions.

10.3 Resources and division of work

The candidate's proposed responsibilities are experimental design, acquisition and analysis, model implementation, uncertainty assessment, cost integration and dissertation synthesis. The supervisor provides scope and scientific oversight. HiLASE access, apparatus ownership and trained operation must be confirmed for each campaign. A named economic-methods consultation should review the decision boundary and allocation/discounting choices before the confirmatory stage.

Laser and high-voltage authorisations, configuration-specific risk assessment, grounding, interlocks and emergency procedures are prerequisites for acquisition. The report is not an operating instruction. CO₂ hardware, pressure approvals and proposed cooperation with Iva Buryšková are follow-on matters and cannot become prerequisites for Tier 1 laboratory access or thesis completion.

11. Risks and scope protection

Risk	Early indication	Response within the dissertation scope
Timing reference dominates the result	Offset or detector uncertainty comparable to the effect	Improve the reference or report a bounded timing claim
Channel descriptor unavailable or unstable	Poor repeatability, early breakdown or missing shot linkage	Narrow the descriptor claim; retain measured operating-domain comparison
Prediction does not transfer	Gain disappears in held-out sessions	Prefer M0, report limits and test robust choices
Costs lack support	No traceable ownership, service or utilisation input	Use declared ranges and break-even conditions; avoid precise savings
No point meets the service requirement	Confidence bounds fail the frozen constraints	Report no feasible solution; quantify the required improvement
Laboratory access slips	Confirmatory slots not reserved	Reduce the number of conditions while preserving independent repetition
Publication review delays eligibility	Core paper not submitted in 2027	Prioritise bounded publishable evidence and address eligibility early
Scope expands to CO ₂ or a demonstrator	New dependencies enter the core schedule	Move them to separately resourced Tier 2/3

Progress reviews should track the evidence remaining for C1, C2 and C3, not only completed activities. A missed gate triggers a scope review and a revised resource plan, not automatic postponement to 2029. Major conflicts in source evidence are recorded and resolved before being converted into thesis claims.

12. Follow-on programme: Tier 2 and Tier 3

Tier 2 asks whether the atmospheric measurement and prediction framework transfers to CO₂ and a selected pressure range. It requires a separate apparatus, gas/pressure safety, calibrated comparison design, access and collaboration agreement. It may use the knowledge gained in Tier 1 but does not change the atmospheric-air dissertation question. Its datasets, papers and medium-transfer claims are not counted toward the mandatory August 2028 deliverables.

Tier 3 may address one application of the resulting knowledge: recovery/history effects, a diagnostic demonstrator or adaptive selection of subsequent shots. Its activation depends on a defined beneficiary, sufficient evidence and separate resources. It need not follow Tier 2 if the selected application uses atmospheric-air results alone; its dependency follows the actual scientific question.

The indicative continuation from autumn 2028 into 2029 is a research outlook. It is not an extended deadline for the dissertation. Earlier completion of an optional study does not make it a new obligation; inclusion of any mature optional result must not displace the writing or publication work required for Tier 1.

13. Proposed dissertation structure and questions for the discussion

The planned monograph contains: (1) problem and technical/economic decision; (2) literature and inherited evidence; (3) apparatus, metrology and data integrity; (4) atmospheric operating domain; (5) predictive model and independent validation; (6) TCO and constrained operating choice; (7) synthesis, limits and follow-on research. Tier 2 and Tier 3 appear as outlooks, not required results chapters.

The opponent and board are asked to assess the following substantive points:

1. Is the combination of C1, C2 and C3 sufficiently original and deep for the registered programme?
2. Is the bounded atmospheric-air configuration family, with one optional robustness comparison, an adequate boundary for the dissertation?
3. Does the chosen channel observable have a defensible physical interpretation and timing relationship to breakdown?
4. Is the distinction between post-pulse prediction and pre-trigger operating choice methodologically sound?
5. Is the proposed laboratory service a meaningful and measurable economic comparison unit?
6. Are the performance requirements, uncertainty treatment and independent evidence sufficient to support an operating recommendation?
7. Can the protected scope and publication route realistically support submission in August 2028?

The requested resolution should record the proposed title, adequacy of the atmospheric core and economic contribution, the evidence standard and the August 2028 submission target. It should explicitly state that Tier 2 and Tier 3 are outside mandatory completion. Formal approval and outstanding study obligations must be recorded by the competent bodies after the discussion.

14. Conclusion

The research foundation consists of seven voltage-reduction series from Master's work in May 2024 and three new doctoral delay series from February 2025. Their publication in ICOLSE and APL establishes continuity and experimental feasibility, but does not yet establish the calibrated predictive and economic contribution proposed here. The disputed monotonic irradiance-delay interpretation is not used.

The next doctoral step is a reproducible atmospheric experiment with a calibrated optical reference, independent validation and a cost model constrained by measured performance. This provides a coherent route from discharge physics to a technical and economic decision while keeping the experimental scope manageable. Tier 1 is designed for submission in August 2028; CO₂ transfer and application development remain separate follow-on research. The professional discussion should test the adequacy and feasibility of this scope rather than require completion of the entire future programme.

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